

Gabriel Franco

665 Commonwealth Ave, Boston, MA

+1 (617) 685-9112 | gvfranco@bu.edu | cs-people.bu.edu/gvfranco

LinkedIn: [gaabrielfranco](#) | GitHub: [gaabrielfranco](#)

Education

Boston University

Sep 2021 – Present

PhD in Computer Science, expected December 2026

Boston, MA

- Advisor: Mark Crovella

Federal University of Viçosa

Aug 2018 – Jun 2021

MSc in Computer Science

Brazil

- Advisor: Giovanni Comarela
- Thesis: New Hyperparameter Strategies for Learning with Label Proportions

Federal University of Viçosa

Mar 2014 – July 2018

B.S. in Computer Science

Brazil

Research Interests

Mechanistic interpretability, representation geometry, feature alignment, and causal circuit discovery. Previous work includes LLM evaluation and learning from label proportions.

Publications

Mechanistic Interpretability

- **Franco, Gabriel**, Carson Loughridge, and Mark Crovella. “Singular Vectors of Attention Heads Align with Features.” <https://arxiv.org/abs/2602.13524>. **Accepted, ICML 2026**.
- **Franco, Gabriel**, Lucas M. Tassis, Azalea Rohr, and Mark Crovella. “Finding Interpretable Prompt-Specific Circuits in Language Models.” <https://arxiv.org/abs/2602.13483>. Under peer review.
- Tang, Zilu, Qiao Zhao, **Gabriel Franco**, Derry Wijaya, Aaron Mueller, Sebastian Schuster, and Najoung Kim. “Do Language Models Track Entities Across State Changes?” **Accepted, ICML 2026**.
- **Franco, Gabriel**, and Mark Crovella. “Pinpointing attention-causal communication in language models.” In The Thirty-ninth Annual Conference on Neural Information Processing Systems. 2025. <https://openreview.net/forum?id=wUoK24u4x7>
- **Franco, Gabriel**, and Mark Crovella. “Sparse Attention Decomposition Applied to Circuit Tracing.” <https://arxiv.org/abs/2410.00340>.

LLM Evaluation

- Calais, Pedro, **Gabriel Franco**, Zilu Tang, Themistoklis Nikas, Wagner Meira Jr., Evimaria Terzi, and Mark Crovella. “Disentangling Text and Math in Word Problems: Evidence for the Bidimensional Structure of Large Language Models’ Reasoning.” Findings of the Association for Computational Linguistics: ACL 2025. 2025. <https://aclanthology.org/2025.findings-acl.656/>

Learning from Label Proportions (LLP)

- **Franco, Gabriel**, Giovanni Comarela, and Mark Crovella. “Evaluating LLP Methods: Challenges and Approaches.” <https://arxiv.org/pdf/2310.19065>.
- **Franco, Gabriel**, Mark Crovella, and Giovanni Comarela. “Dependence and Model Selection in LLP: The Problem of Variants.” Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. 2023.

Other Publications

- Ghaemi, Golsana, **Gabriel Franco**, Kazem Taram, and Renato Mancuso. “MEMSCOPE: Open-Source Kernel-Level Framework for Heterogeneous Memory Characterization.” In 2025 IEEE Real-Time Systems Symposium (RTSS), pp. 500-513. IEEE, 2025.
- **Franco, Gabriel**, Marcos Henrique Fonseca Ribeiro, and Giovanni Comarella. “Towards an interpretable metric for DOTA 2 players: An unsupervised learning approach.” 2019 8th Brazilian Conference on Intelligent Systems (BRACIS). IEEE, 2019.
- Gomes, Andréia Patricia, Brenda Silveira Valles Moreira, Felipe José Dutra Dias, Victor Hiroshi Bastos Inoue, **Gabriel Vita Silva Franco**, et al. “Plasmodium Falciparum Infection: In Silico Preliminary Studies.” Abakós 5.1 (2016): 63-83.

Invited Talks

“Understanding What Happens Inside LLMs: Mechanistic Interpretability and How Attention Works.”

- Universidade Federal de Viçosa, April 4, 2026.

“Understanding Attention from First Principles: From Singular Vectors to Interpretable Causal Circuits.”

- Brown University, March 12, 2026.
- TINLab, Boston University, March 6, 2026.

“Pinpointing Attention-Causal Communication in Language Models.”

- BAAIGL Lab, Boston University, September 24, 2025.

Research Experience

Research Assistant

Sep 2021 – Present

Boston University

Boston, MA

- Towards understanding internal communication mechanisms in Large Language Models (LLMs): Developed a novel spectral decomposition method to identify causal communication circuits in LLMs, enabling new applications in circuit tracing and low-rank model interventions.
- Hyperparameter selection strategies and new datasets for Learning from Label Proportions (LLP): Proposing that LLP problems can be characterized into distinct variants, showing that different hyperparameter selection strategies are needed for each, and proposing a new set of benchmark datasets to account for these variants.

Master’s Student Researcher

Aug 2018 – Jul 2020

CAPES (Brazilian Government Research Funding Agency)

Brazil

- New Hyperparameter Strategies for Learning with Label Proportions (LLP): Worked on improving the cross-validation strategies for LLP.

Undergraduate Student Researcher

Mar 2015 – Jul 2018

CNPq/FAPEMIG (Brazilian Government Research Funding Agencies)

Brazil

- Developed the Bio-Oracle software: Bio-ORACLE is a software written in JAVA which uses data mining techniques to help decision-making in bioethics. Bio-ORACLE was developed at the Laboratory of Epidemiological and Computational Methods in Health of Department of Medicine and Nursing in Federal University of Viçosa.
- Implementing malaria and plasmodium on Autosimmune: Autosimmune is a multiagent human immune system simulator made in JAVA. Autosimmune was developed at the Laboratory of Epidemiological and Computational Methods in Health of Department of Medicine and Nursing in Federal University of Viçosa.

Industry Experience

Data Scientist Intern

May 2024 – Aug 2024

Microsoft

Redmond, WA

- Successfully fine-tuned and optimized multi-modal SLMs for local image tagging, achieving ~85% of a large foundation model's performance while meeting <5s inference targets.

Data Scientist

Sep 2020 – Jul 2021

SEEK

Brazil (Remote)

- Optimized a large-scale recommender system, reducing average response time by over 50% to support scalability for a growing user base.
- Designed and A/B tested a new job recommendation model, increasing click-through rate by 5% and job applications by 10%.

Data Scientist

Jul 2020 – Sep 2020

Localiza

Brazil (Remote)

- Developed and validated a machine learning model to identify reliable customers; presented feasibility analysis and results to stakeholders to inform business decision-making.

Book Chapters & Educational Materials

- Siqueira-Batista, Rodrigo, et al. Parasitologia: Fundamentos e Prática Clínica. Guanabara, 2020. ISBN 9788527735735. (I co-wrote Chapter 4 about the computational approach in the study of parasitic diseases)
- Comarela, G.; Franco, G.; Trois, C.; Liberato, A.; Martinello, M.; Corrêa, J. H.; Villça, R. Introdução à Ciência de Dados: Uma Visão Pragmática utilizando Python, Aplicações e Oportunidades em Redes de Computadores SBRC 2019 (Short Course)

Teaching Experience

Teaching Assistant

Jan 2025 – May 2025

Boston University

Boston, MA

- CAS CS 132: Geometric Algorithms.

Teaching Assistant

Sep 2023 – Dec 2023

Boston University

Boston, MA

- CDS DS 701: Tools for Data Science.

Instructor

Nov 2019

Federal University of Viçosa

Brazil

- L^AT_EX short course.

Teaching Assistant

Mar 2019 – Jul 2019

Federal University of Viçosa

Brazil

- INF 100 - Introduction to Programming I.

Academic Service

- Reviewer COLM 2026
- Reviewer ICML 2026 (recognized as Gold Reviewer)
- Reviewer ACL ARR 2026 January
- Reviewer ICLR 2026
- Reviewer Mechanistic Interpretability Workshop at NeurIPS 2025
- Reviewer NeurIPS 2025

- Reviewer ICLR 2025

Awards & Grants

- NeurIPS 2025 Scholar Award
- KDD'23 Student Travel Award
- KDD'22 Student Travel Award
- CNPq Research Scholarship (PIBIC/CNPq) - 2018
- CNPq Research Scholarship (PIBIC/CNPq) - 2017
- CNPq Research Scholarship (PIBIC/CNPq) - 2015
- FAPEMIG Research Scholarship (PIBIC/FAPEMIG) - 2015

Technical Skills

Languages	Python, C/C++, Java.
ML/DS Libraries	PyTorch, Transformers, PEFT, Scikit-learn, Pandas, Numpy, Matplotlib, Seaborn, TransformerLens, nnsight.
Other Tools	Git, Linux, SQL, Redis, ElasticSearch.

Relevant Coursework

- CS542 - Machine Learning
- CS565 - Algorithmic Data Mining
- DS563 - Algorithmic Techniques for Taming Big Data
- CS505 - Introduction to Natural Language Processing
- LX690 - Metrics and Evaluation in Natural Language Processing (Audited)

Languages

- Portuguese (Native or bilingual proficiency)
- English (Full professional proficiency)
- Spanish (Elementary proficiency)

Additional Activities

- ACM International Collegiate Programming Contest - Regionals - 2014, 2015, 2017
- Minas Gerais State Programming Contest - 2014, 2015, 2017

References

Prof. Mark Crovella Boston University	crovella@bu.edu
Prof. Evimaria Terzi Boston University	evimaria@bu.edu
Prof. Aaron Mueller Boston University	amueller@bu.edu

Prof. Najoung Kim

Boston University

najoung@bu.edu

Prof. Giovanni Ventrone Comarella

Universidade Federal do Espírito Santo

gc@inf.ufes.br